AI-LAB

AI-LAB · OPERATIONS

Operations Playbook

Run it locally on Docker, keep it healthy, and run every test suite.

Version 1.0 · 10 November 2026 · For the operator and the release engineer

What you need

Docker Engine 27+ and Docker Compose 2.30+. Make and Git. At least 4 vCPU, 8 GB RAM and 20 GB free disk. Outbound HTTPS to DeepSeek, GitHub and Docker Hub.

Docker only. Rust, Node/npm, Python and OpenSSL run in the pinned containers. Do not install or use host toolchains; use the Make targets and the **scripts/run-*.sh** wrappers (ADR-0027).

First start

```
git clone <repository> ai-lab
cd ai-lab
make setup
make generate-key
make doctor
make up
make seed-demo
```

Open **http://127.0.0.1:8080**. The core profile starts Postgres, Redis, migrate, api, worker, runner, the Docker socket proxy and web. The demo seed adds one lab, four agents, one project, one SOW, one plan, 12 cards and 30 events.

Run the doctor first. It checks Docker, Compose, images, ports, disk, **.env** keys and connectivity. Fix every hard failure before **make up**.

Daily commands

Command	What it does
make ps	Container status and health.
make logs	Follow the core logs.
make up / make down	Start or stop; volumes are preserved.
make reset	Stop, remove volumes, migrate and seed demo. Destructive.
make backup	Dump the database and write an artifact manifest.
make restore-drill BACKUP=...	Restore into a scratch database and compare row counts.

Do not reset blindly. make reset deletes local data. Run **make backup** first if you care about it.

Safety defaults. The app binds to loopback. Never commit .env. Keep the per-card and per-day budgets as hard stops. Stop the stack with **make down** when you are done.

Run the tests

Command	What it covers
make check	Fast local gate: format, lint, typecheck, unit tests, docs, fitness and Compose.
make test	Backend unit and integration tests in Docker.
make frontend-test	Frontend unit tests in Docker.
make frontend-lint && make frontend-build	Frontend quality and bundle budgets in Docker.
make e2e	Playwright end-to-end suite.
make bdd-list	List the six BDD suites.
make bdd	Run all 480 BDD scenarios.
make bdd SUITE=core	Run one BDD suite.
make bdd UP_TO=delivery	Run the ordered prefix up to delivery.
make bdd-set-api-key	Load the live DeepSeek key for live runs.

The six BDD suites

#	Suite	Scenarios
1	core	41
2	kanban	50
3	delivery	50
4	platform	74
5	office	137
6	management	128
	Total	480

Only **make bdd** is supported. The preflight reads **.docs/e2e/req.md** and blocks the run if a manual item is missing. The full output is written to **web/e2e-bdd/reports/make-bdd.log**.

Live vs fake. Automated runs use the fake LLM and SCM harness. The live DeepSeek key is loaded only for **@live** scenarios through **make bdd-set-api-key**; it is never written to .env or a report. Provider persistence is owned by WP-07/WP-16; until it lands, the live key loader reports HTTP 501. Use the default fake-harness suite.

Troubleshooting

Symptom	First move
Port 8080 is busy	Run make doctor and stop the conflicting process.
migrate exits non-zero	Read make logs, fix the migration, run make up again.
Web returns 502	Run make ps and check the api logs.
BDD will not start	Complete .docs/e2e/req.md and read reports/make-bdd.log.
Unknown suite	Run make bdd-list and use one of the six ids.
Disk is full	Run make clean and prune worktrees or artifacts.